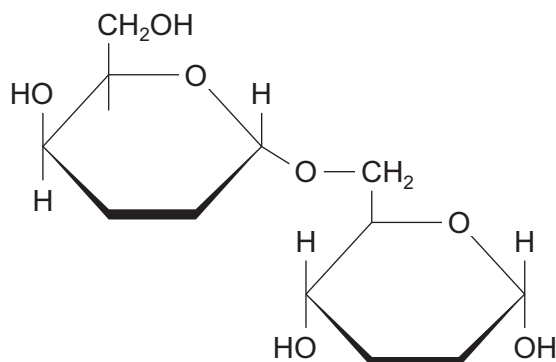


5. Melibiose is a reducing sugar that is formed from galactose and glucose and is thirty times sweeter than sucrose.

(a) The diagram below shows the structure of melibiose.



- (i) **Complete the diagram above** to show how this molecule would be broken down into two monosaccharides. [2]
- (ii) State the type of reaction that occurs when this molecule is broken down into two monosaccharides. [1]

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- (iii) Describe a biochemical test that could be used to determine the presence of a reducing sugar such as melibiose. [2]

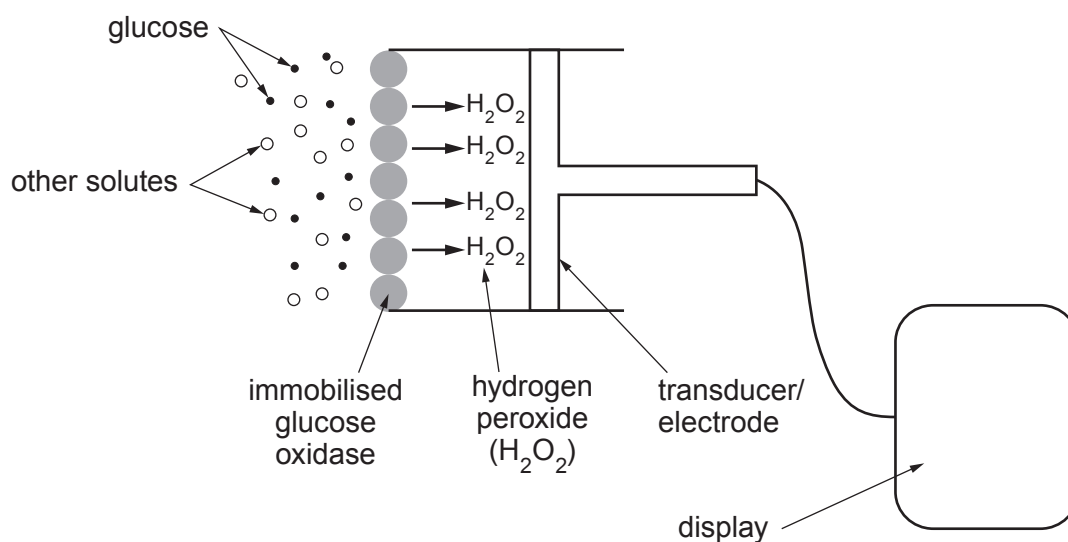
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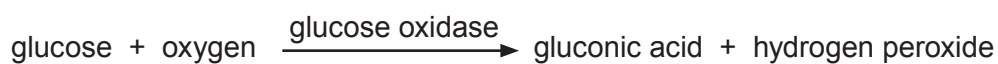
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- (b) Biosensors make use of immobilised enzymes to detect specific molecules in a mixture. The diagram below shows a possible structure of a biosensor that could be used to detect the presence of glucose in a urine sample.



Within the biosensor, the glucose oxidase catalyses the following reaction:



The concentration of hydrogen peroxide is detected by the transducer and converted into an electrical signal, shown as a measurement on the display.



- (i) Using your knowledge of enzymes explain why this biosensor will only detect the presence of glucose. [3]

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- (ii) With reference to the information given, explain how the biosensor could be used to give a measurement of the glucose concentration. [2]

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- O[C@H]1O[C@H](O)[C@H](O)[C@@H](O)[C@H]1O

Experiments have shown that arabinose can inhibit glucose oxidase and therefore affect the accuracy of the biosensor. Using the information provided in part (a) and the diagram above, explain why the contamination of a urine sample with arabinose would affect the accuracy of the biosensor.

[4]

